

Household Carbon Footprint & Target Tracker

Use-Case Flow Specification

Use Case: Log Monthly Consumption

Actor: Resident

Preconditions

- Resident is authenticated and logged into the system.
- A household profile already exists in the system.

Postconditions

- A new monthly log entry is stored for the household.
- CO2e emissions are computed and linked to that entry.
- Progress toward the household's active reduction target is recalculated.

Main Success Scenario

1. Resident navigates to "Log this month's data."
2. System presents a form for electricity (kWh), gas usage, and commute distance (km).
3. Resident enters values for each field and submits the form.
4. System validates that all inputs are present and non-negative numbers.
5. System invokes the Compute CO2e Emissions use case (<<include>>) using EPA conversion factors.
6. System stores the log entry together with the computed CO2e value.
7. System recalculates and updates the Resident's progress toward their active reduction target.
8. System displays a confirmation screen showing the computed emissions figure.

Alternate Flow – A1: Invalid Input

- 3a. Resident submits the form with a missing or negative value.
- 3a1. System displays an inline validation error identifying the offending field.
- 3a2. System does not save the log entry.
- 3a3. Flow returns to step 3, awaiting corrected input from the Resident.